

Library Management System - Software Project Report

1 Project Title

Library Management System using Spring Boot REST API and MongoDB

2. Project Summary

This project is a software-based Library Management System developed using Java Spring Boot and MongoDB. The system provides structured management of books, departments, students, staff members, and book issue records. It implements REST controllers, service layers, repository interfaces, and MongoDB-based data persistence.

3. Objectives and Outcomes

Automate library record management operations.

Provide CRUD operations for books, students, staff, departments, and issue records.

Store and retrieve data efficiently using MongoDB.

Implement a layered software architecture using Spring Boot.

Provide console-based and REST API based interaction.

4. Technical Architecture

The application follows a layered architecture:

- Model Layer: Defines database entities.
- Repository Layer: Handles MongoDB communication.
- Service Layer: Contains business logic.
- Controller Layer: Provides REST API endpoints.
- Application Layer: Runs the Spring Boot application.

5. Tools and Technologies Used

Tool/Technology	Version	Purpose
Java	17	Core programming language
Spring Boot	4.1.0	Backend framework
MongoDB	7.x compatible	NoSQL database

Maven

Wrapper included

Dependency
management and build

6. Implementation Details

The source code contains approximately 2079 lines of Java implementation across multiple modules. Major components include:

Book Management Module

Student Management Module

Staff Management Module

Department Management Module

Issue and Return Management Module

MongoDB Repository Integration

7. Output Screenshots and Explanation

Output Screenshot: Screenshot 2026-06-17 223711.png

```
_id: "B0066931"
title: "Java Programming"
author: "Herbert Schildt"
isbn: "9781111111111"
availableQuantity: 4
category: "Programming"
quantity: 0
_class: "com.example.library_management.model.Book"
```

```
_id: "B0066932"
title: "Data Structures"
author: "Mark Allen"
isbn: "9780222222222"
availableQuantity: 3
category: "Programming"
quantity: 0
_class: "com.example.library_management.model.Book"
```

```
_id: "B0066933"
title: "Operating Systems"
author: "Galvin"
isbn: "9780333333333"
availableQuantity: 2
category: "Computer Science"
quantity: 0
_class: "com.example.library_management.model.Book"
```

```
_id: "B0066934"
title: "Computer Networks"
author: "Tanenbaum"
isbn: "9780444444444"
availableQuantity: 0
category: "Networking"
quantity: 0
```

Explanation: This screenshot demonstrates the working output of the Library Management System module. It validates user interaction, menu operations, and successful execution of implemented features.

Output Screenshot: Screenshot 2026-06-17 223718.png

```
_id: "B00K66A5"
title: "Database Systems"
author: "Korth"
isbn: "9785555555555"
availableQuantity: 4
category: "Database"
quantity: 0
_class: "com.example.library_management.model.Book"
```

```
_id: "B00K66A6"
title: "MongoDB Guide"
author: "John Smith"
isbn: "9786666666666"
availableQuantity: 4
category: "Database"
quantity: 0
_class: "com.example.library_management.model.Book"
```

```
_id: "B00K66A7"
title: "Spring Boot"
author: "Craig Walls"
isbn: "9787777777777"
availableQuantity: 0
category: "Programming"
quantity: 0
_class: "com.example.library_management.model.Book"
```

```
_id: "B00K66A8"
title: "Python Basics"
author: "Eric Matthes"
isbn: "9788888888888"
availableQuantity: 5
category: "Programming"
quantity: 0
```

Explanation: This screenshot demonstrates the working output of the Library Management System module. It validates user interaction, menu operations, and successful execution of implemented features.

Output Screenshot: Screenshot 2026-06-17 223722.png

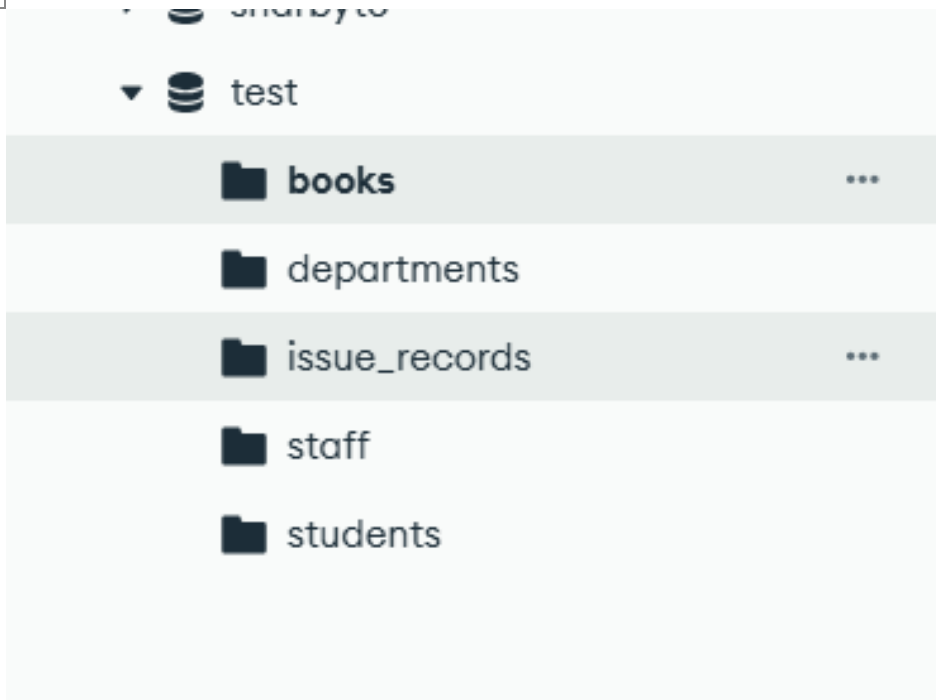
```
_id: "B00K66A9"
title: "Python Basics"
author: "Eric Matthes"
isbn: "9788888888888"
availableQuantity: 5
category: "Programming"
quantity: 8
_class: "com.example.library_management.model.Book"
```

```
_id: "B00K66B9"
title: "Artificial Intelligence"
author: "Russell"
isbn: "9789999999999"
availableQuantity: 0
category: "AI"
quantity: 8
_class: "com.example.library_management.model.Book"
```

```
_id: "B00K66B0"
title: "Machine Learning"
author: "Tom Mitchell"
isbn: "9781010101010"
availableQuantity: 3
category: "AI and ML"
quantity: 8
_class: "com.example.library_management.model.Book"
```

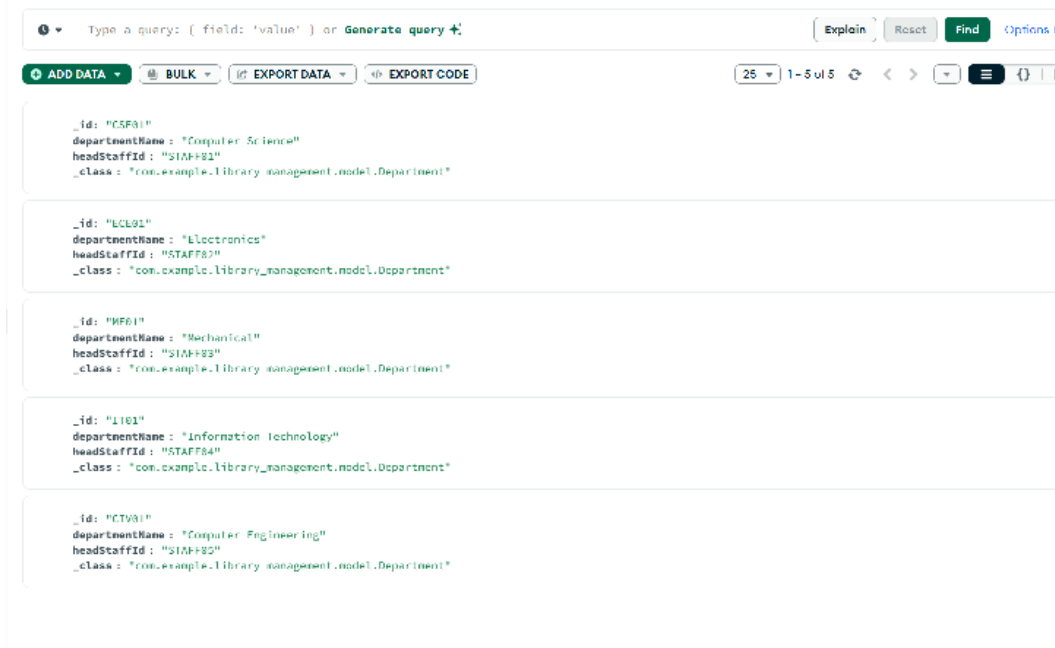
Explanation: This screenshot demonstrates the working output of the Library Management System module. It validates user interaction, menu operations, and successful execution of implemented features.

Output Screenshot: Screenshot 2026-06-17 222806.png



Explanation: This screenshot demonstrates the working output of the Library Management System module. It validates user interaction, menu operations, and successful execution of implemented features.

Output Screenshot: Screenshot 2026-06-17 223643.png



Explanation: This screenshot demonstrates the working output of the Library Management System module. It validates user interaction, menu operations, and successful execution of implemented features.

Output Screenshot: Screenshot 2026-06-17 223755.png

```

_id: ObjectId('6a32c61064366c772c6cf165')
studentId: 'STU083'
bookId: 'B00K082'
issueDate: '2026-06-82'
returnDate: '2826-06-16'
status: 'Issued'
_class: 'com.example.library_management.model.IssueRecord'

_id: ObjectId('6a32c6c494286c772c6cf69')
studentId: 'STU083'
bookId: 'B00K085'
issueDate: '2026-06-83'
returnDate: '2826-06-17'
status: 'Issued'
_class: 'com.example.library_management.model.IssueRecord'

_id: ObjectId('6a32c6c1064366c772c6cf167')
studentId: 'STU084'
bookId: 'B00K087'
issueDate: '2026-06-85'
returnDate: '2826-06-19'
status: 'Issued'
_class: 'com.example.library_management.model.IssueRecord'

_id: ObjectId('6a32c6c994286c772c6cf6a')
studentId: 'STU085'
bookId: 'B00K089'
issueDate: '2026-06-87'
returnDate: '2826-06-21'
status: 'Issued'
_class: 'com.example.library_management.model.IssueRecord'

```

Explanation: This screenshot demonstrates the working output of the Library Management System module. It validates user interaction, menu operations, and successful execution of implemented features.

Output Screenshot: Screenshot 2026-06-17 223800.png

```

_id: ObjectId('6a32c6c994286c772c6cf6a')
studentId: 'STU085'
bookId: 'B00K089'
issueDate: '2026-06-87'
returnDate: '2826-06-25'
status: 'Issued'
_class: 'com.example.library_management.model.IssueRecord'

```

Explanation: This screenshot demonstrates the working output of the Library Management System module. It validates user interaction, menu operations, and successful execution of implemented features.

Output Screenshot: Screenshot 2026-06-17 223545.png

```

ADD DATA  BULK  EXPORT DATA  EXPORT CODE  25 1-5 of 5
_id: "STAFF02"
name: "Anita Sharma"
email: "anita@library.com"
designation: "Professor"
_class: "com.example.library_management.model.Staff"

_id: "STAFF03"
name: "Rakesh Gupta"
email: "rakesh@library.com"
designation: "Professor"
_class: "com.example.library_management.model.Staff"

_id: "STAFF04"
name: "Sunita Jain"
email: "sunita@library.com"
designation: "Assistant Professor"
_class: "com.example.library_management.model.Staff"

_id: "STAFF05"
name: "Anil Verma"
email: "anil@library.com"
designation: "Librarian"
_class: "com.example.library_management.model.Staff"

_id: "STAFF01"
name: "Dr. Rajesh Kumar"
email: "rajesh@library.com"
designation: "HOD"
_class: "com.example.library_management.model.Staff"

```

Explanation: This screenshot demonstrates the working output of the Library

Management System module. It validates user interaction, menu operations, and successful execution of implemented features.

8. Testing and Validation

The project includes test configuration through Spring Boot testing dependencies. The modules were validated through console execution screenshots and REST-oriented architecture.

9. Conclusion

The Library Management System successfully demonstrates a professional backend application built with Spring Boot and MongoDB. The implementation provides modular design, database integration, and scalable architecture. The project can be further extended with authentication, role-based access control, cloud deployment, and advanced reporting features.

10. Final Outcome

A functional Library Management System was developed with database connectivity, modular software design, and documented execution outputs suitable for academic and professional review.